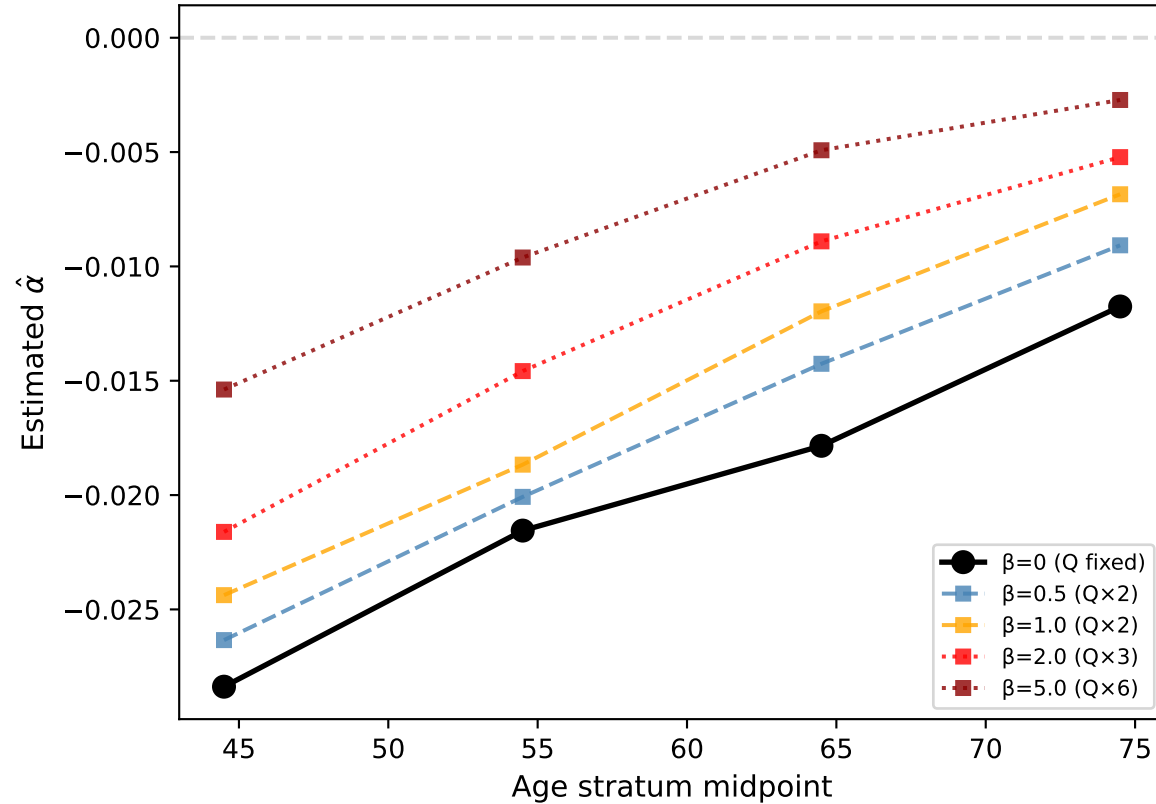
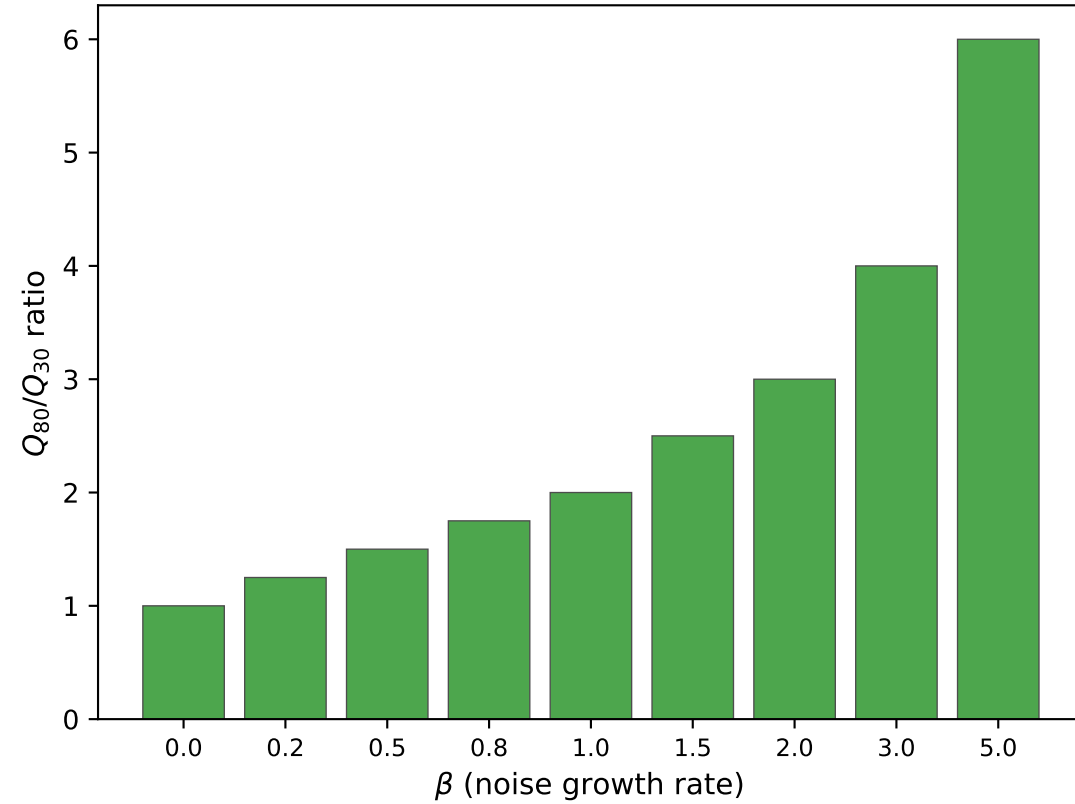


Diffusion Covariance (Q) Sensitivity Analysis

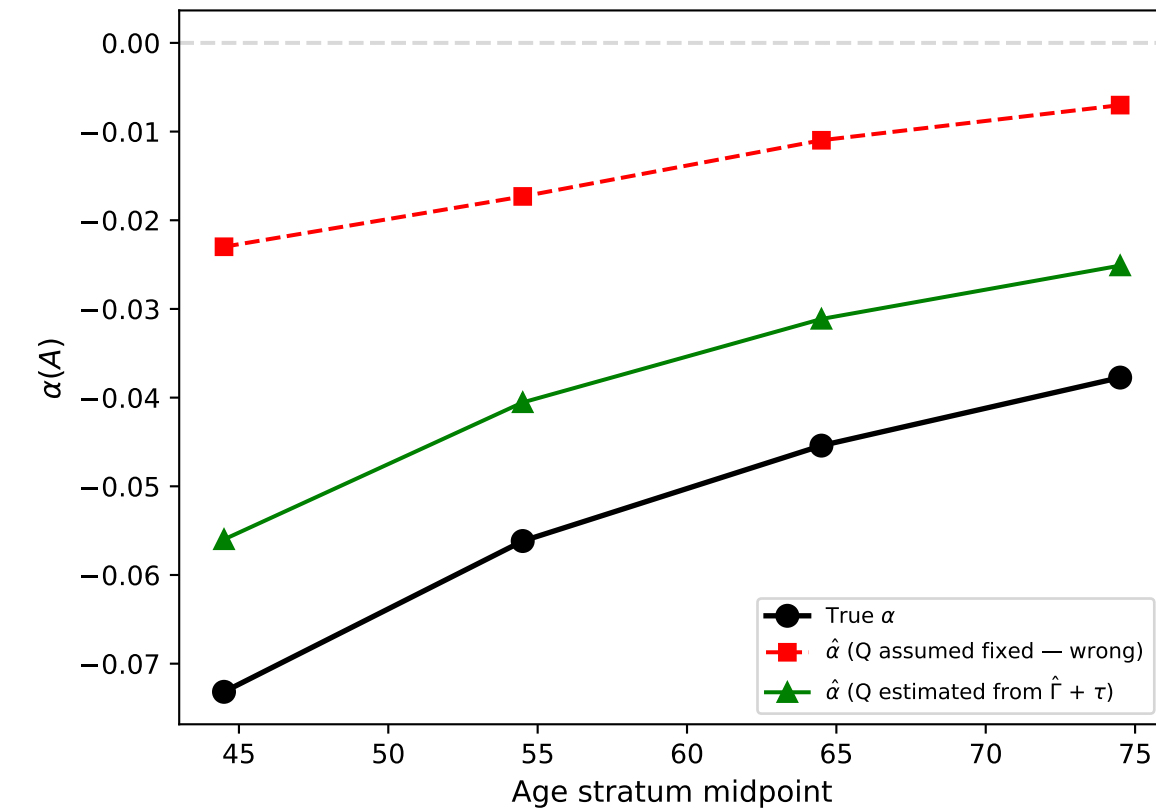
(a) $\hat{\alpha}$ trend when Q assumed fixed but true Q varies



(b) Does monotone $\hat{\alpha}$ trend survive? (green=yes)



(c) Q-estimation corrects the $\hat{\alpha}$ pipeline



Q-SENSITIVITY VERDICT

Threshold $\beta^* \approx >5.0$
(Q at age 80 = $>6 \times$ Q at age 30)

If noise increases by less than $>5 \times$ over the lifespan, the monotone $\hat{\alpha}$ trend reflects genuine stability-margin erosion, not artifact of increasing noise.

MITIGATION STRATEGIES:

- Estimate Q from $\hat{\Gamma} + \text{literature } \tau$ $\rightarrow$ corrects bias even when Q varies
- Test-retest q from within-individual short-term variability (CGM, serial CRP)
- Report sensitivity across $\beta = [0, 2]$ as standard robustness check

R4 NOTE: $\lambda_{\max}(\hat{\Gamma})$ does not require Q specification — robust to Q misspecification by construction.